VERIDIAN PHARMA LTD. — Quality & Compliance Directorate, Hyderabad Facility

Standard Operating Procedure: Sample Storage and Handling

Document ID: SOP-LAB-007	**Version:** 1.0	**Effective Date:** 2026-05-20
Owner: Arjun Nair (Lab Manager)	**Department:** Laboratory Operations	**Status:** Effective

1. Purpose

To establish rigid and standardized operational protocols for receiving, accessioning, cataloging, storing, and transferring analytical and stability test samples across all QC laboratories at Veridian Pharma Ltd., Hyderabad.

Maintaining absolute environmental integrity during sample handling is vital to ensure that analytical test results accurately reflect product specifications and are free from artifactual degradation, contamination, or container compromise.

This procedure ensures alignment with pharmacopoeial standards and Good Laboratory Practice (GLP) expectations.

2. Scope

Applies to all analytical chemistry, wet chemistry, microbiological, packaging inspection, and stability testing laboratories operating under the Laboratory Operations Directorate.

Covers all incoming active pharmaceutical ingredients (APIs), excipients, primary packaging materials, in-process bulk solutions, finished dosage forms, and retention samples.

Personnel including sample receipt clerks, QC analysts, microbiologists, and laboratory managers are strictly bound to comply with this procedure.

3. Responsibilities

• **Sample Receipt Custodian:** Inspect physical containers, verify shipping paperwork, check temperature indicators upon receipt, and log specimens into LIMS.

• **Quality Control Analyst:** Verify sample identification prior to withdrawal, adhere to sampling aliquoting instructions, and return unused portions to designated chambers.

• **Lab Manager (Arjun Nair):** Maintain environmental compliance, review chamber temperature monitoring logs, oversee inventory reconciliations, and direct out-of-spec environmental investigations.

• **Quality Assurance Auditor:** Perform periodic independent inspections of sample storage facilities, review logbooks, and verify adherence to storage temperature limits.

4. Procedure

4.1 Receipt and Verification of Primary Packaging: Upon receipt, analysts inspect the external container for seal integrity and appropriate labeling to identify any shipping damage or container breaches.

4.2 Primary Refrigerated Storage Condition: Samples stored at 2-8 °C.

4.3 Analytical Testing Storage Window: Max 48 hours in storage before analysis.

4.4 Mandatory Register Logging: Every entry logged in the Storage Register (form LAB-SR-02).

4.5 Frozen Sample Preservation: Freezer samples at -20 °C.

4.6 Continuous Environmental Telemetry: Storage chambers must be monitored continuously with calibrated temperature sensors connected to the automated facility monitoring system.

4.7 Excursion Escalation Workflow: In the event of temperature excursions outside designated limits, notify the Lab Manager immediately to initiate containment and stability impact reviews.

4.8 Aliquoting and Thermal Cycling Minimization: Analysts must withdraw only the requisite aliquot for analysis to prevent unnecessary thermal cycling that could accelerate chemical degradation.

4.9 Tamper-Evident Long-Term Storage: Retest samples and reserve samples must be sealed with tamper-evident tape prior to placement in long-term storage.

4.10 Final Disposal and Inventory Reconciliation: Disposed samples must be de-logged from the inventory system in accordance with waste disposal protocols to maintain an accurate real-time inventory ledger.

5. Records and Documentation

Storage Register form LAB-SR-02, continuous chamber temperature traces, excursion event files, and custody transfer registers must be retained for audit examination.

Records must be kept for a minimum retention duration in line with master document retention rules.

6. Revision History

Version	Effective Date	Change Description
1.0	2026-05-20	Initial release for laboratory sample storage and handling.